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Probability and Random variable

Basic concepts of Probability:

Definition of Probability:

If there are n equally likely mutually exclusive and exhaustive outcomes (m) and are favorable to an event a, probability of total number of events of a, then

$$p(a) = \frac{\text{no. of favorable cases}}{\text{total no. of events}}$$

Note:

1. $P(a) = 1$

   The event a is called certain event

2. $P(a) = 0$

   The event a is called impossible event

3. $P(a) + \overline{P(a)} = 1$

   Total probability $= 1$

① An integer is chosen from 2 to 15. what is probability that it is prime?

soln:-

n = Total no = 14

m = favorable cases = 2, 3, 5, 7, 11, 13 = 6

$$P(A) = \frac{m}{n} = \frac{6}{14} = \frac{3}{7}$$

③ Two dies are thrown. Find the probability that

i.) The total ~~no~~ of the numbers on the top faces is 9.

ii) The top face numbers are same

iii) The sum of the numbers on the top face is less than seven.

Soln:-

$$S = \begin{Bmatrix} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6) \\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) & (2,6) \\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) & (3,6) \\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & (4,6) \\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) & (5,6) \\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6) \end{Bmatrix}$$

$n(S) = 36$

i.) $A = \{(3,6), (4,5), (5,4), (6,3)\}$

$n(A) = 4$

$$P(A) = \frac{n(A)}{n(S)} = \frac{4}{36} = \frac{1}{9}$$

ii) $B = \{(1,1), (2,2), (3,3), (4,4)\ (5,5)\ (6,6)\}$

$n(B) = 6$

$$P(B) = \frac{6}{36} = \frac{1}{6}$$

iii)
$$C = \begin{Bmatrix} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) \\ (2,1) & (2,2) & (2,3) & (2,4) \\ (3,1) & (3,2) & (3,3) \\ (4,1) & (4,2) \\ (5,1) \end{Bmatrix}$$

$n(C) = 15$

$$P(C) = \frac{15}{36} = \frac{5}{12}$$

③ Out of a sample of 200 items. 20 items are found to be defective. Find the probability that an item chosen at random from the sample is not defective.

soln:-

$$P(A) = \frac{20}{200} = \frac{1}{10}$$

$$P(A) + P(\overline{A}) = 1$$

$$P(A) = 1 - P(\overline{A})$$

$$= 1 - \frac{1}{10}$$

$$P(A) = \frac{9}{10}$$

④ What is the change that a leaf year selected at random will contains 53 wednesday.

soln:-

Total days = 366 = 7 (52) + 2

$$7 \overline{)366} \quad 52$$
$$35$$
$$16$$
$$14$$
$$2$$

$S = \{SM, MT, TW, WT, TF, FS, SS\}$

$n(S) = 7$ $\quad$ $n(A) = 2$

$$P(A) = \frac{2}{7}$$

⑤ A bag contains four red, 5 white and 6 Black balls. What is the probability that two balls drawn are red and black

soln:-

Total no. of balls = 15

P(two balls drawn are Red and Black) $= \frac{4C_1 \times 6C_1}{15C_2}$

$$= \frac{\frac{4 \times 3}{1 \times 2} \times \frac{6}{1}}{\frac{15 \times 14}{1 \times 2}}$$

$$= \frac{4 \times 6}{15 \times 7} \qquad \text{(or) } nC_r = \frac{n!}{(n-r)!\,r!}$$

$$P(A) = \frac{24}{105}$$

Axioms of probability (or) Properties (or) Conditions:

i) $0 \leq P(A) \leq 1$

ii) $P(S) = 1$

iii) $P\left(\cup A_n\right) = \sum_{n=1}^{\infty} P(A_n)$

Discrete and continuous Random variable:

1. Discrete Random variable (Probability mass function):

$P(X=x) = P_i$

i) $\Sigma P_i = 1$

For a discrete Random variable X taking atmost a countably infinite no of values the probability mass function is defined as

2. Continuous Random variable (Probability density function):

$f(X=x) = f(x) \qquad a \leq x \leq b$

i) $\int_a^b f(x)\,dx = 1$

The probability density function $f(x)$ of a continuous ramdom variable X is defined as such that i) $\int_{-\infty}^{\infty} f(x)\,dx = 1$, ii) $f(x) \geq 0$

Cumulative Distribution function (or) Distribution function:

$$F(x) = P(X \leq x)$$

$$\boxed{F'(x) = f(x)}$$

(1) A discrete random variable X has the probability following distribution

Value of $X = x$	0	1	2	3	4	5	6	7	8
$P(X=x)$	a	$3a$	$5a$	$7a$	$9a$	$11a$	$13a$	$15a$	$17a$

i) Find the values of a

ii) Find $P(X<3)$, $P(0<X<3)$, $P(X \geq 3)$

iii) Find Distributive function of X $(F(x))$

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Soln:-

i) Find 'a':

$$\Sigma P_i = 1$$

$$a+3a+5a+7a+9a+11a+13a+15a+17a=1$$

$$81a = 1$$

$$a = \frac{1}{81}$$

$X=x$:	0	1	2	3	4	5	6	7	8
$P(X=x)$:	$\frac{1}{81}$	$\frac{3}{81}$	$\frac{5}{81}$	$\frac{7}{81}$	$\frac{9}{81}$	$\frac{11}{81}$	$\frac{13}{81}$	$\frac{15}{81}$	$\frac{17}{81}$

ii)

i) $P(X<3) = P(X=0)+P(X=1)+P(X=2)$

$$= \frac{1}{81} + \frac{3}{81} + \frac{5}{81}$$

$$= \frac{9}{81}$$

$$= \frac{1}{9}$$

ii) $P(0<X<3) = P(X=1)+P(X=2)$

$$= \frac{3}{81} + \frac{5}{81}$$

$$= \frac{8}{81}$$

iii.) $P(X \geq 3) = 1 - P(X < 3)$

$= 1 - \left\{\frac{1}{9}\right\}$

$= \frac{8}{9}$

iv.) Cumulative Distribution function.

$F(x) = P(X \leq x)$

$$F(x) = \begin{cases} 0 & , -\infty < x < 0 \\ \frac{1}{81} & , 0 \leq x < 1 \quad (X<1) \\ \frac{4}{81} & , 1 \leq x < 2 \quad (X<2) \\ \frac{9}{81} & , 2 \leq x < 3 \quad (X<3) \\ \frac{16}{81} & , 3 \leq x < 4 \quad (X<4) \\ \frac{25}{81} & , 4 \leq x < 5 \quad (X<5) \\ \frac{36}{81} & , 5 \leq x < 6 \quad (X<6) \\ 49/81 & , 6 \leq x < 7 \quad (X<7) \\ 64/81 & , 7 \leq x < 8 \quad (X<8) \\ 1 & , 8 \leq x < \infty \quad (X<9) \end{cases}$$

⑤ A random variable X has a following ~~production~~ probability ~~production~~ distribution

x :	-2	-1	0	1	2	3
$P(x)$:	0.1	k	0.2	$2k$	0.3	$3k$

i.) Find k

ii.) Evaluate $P(X<2)$ and $P(-2<X<2)$

iii.) Find cummulative distributive of X

iv.) Evaluate mean of X.

soln:

i) find k:

$\Sigma P_i = 1$

$6k + 0.6 = 1$

$6k = 1 - 0.6$

$6k = 0.4$

$k = \frac{0.4}{6}$

$k = \frac{4}{60} = \frac{1}{15}$

$\boxed{k = \frac{1}{15}}$

x :	-2	-1	0	1	2	3
$P(x)$:	$\frac{1}{10}$	$\frac{1}{15}$	$\frac{2}{10}$	$\frac{2}{15}$	$\frac{3}{10}$	$\frac{3}{15}$

ii) $P(X < 2) = 1 - P(X \geq 2)$

$= 1 - \left(\frac{3}{10} + \frac{3}{15}\right)$

$= 1 - \left(\frac{9+6}{30}\right)$

$= 1 - \left(\frac{15}{30}\right)$

$= 1 - \frac{1}{2}$

$P(X < 2) = \frac{1}{2}$

$P(-2 < X < 2) = \frac{1}{15} + \frac{2}{10} + \frac{2}{15}$

$= \frac{2 + 6 + 4}{30}$

$= \frac{12}{30}$

$P(-2 < X < 2) = \frac{2}{5}$

iii)

$$F(x) = \begin{cases} 0, & -\infty < x < -2 \\ \frac{1}{10}, & -2 \le x < -1 \quad (x < -1) \\ \frac{1}{6}, & -1 \le x < 0 \quad (x < 0) \\ \frac{11}{30}, & 0 \le x < 1 \quad (x < 1) \\ \frac{1}{2}, & 1 \le x < 2 \quad (x < 2) \\ \frac{4}{5}, & 2 \le x < 3 \quad (x < 3) \\ 1, & 3 \le x < \infty \quad (x < \infty) \end{cases}$$

iv) <u>Mean:</u>

$$E(x) = \Sigma x P(X = x) = \Sigma x P_i$$

Variance:

$$Var(x) = E(x^2) - (E(x))^2$$

$$E(x^2) = \Sigma x^2 P(x = x)$$

$$E(x) = \Sigma x_i P_i \quad \text{(discrete)}$$

$$-2 \times \frac{1}{15} + -1 \times \frac{1}{15} + 0 \times \frac{2}{10} + 1 \times \frac{2}{15} + 2 \times \frac{3}{10} + 3 \times \frac{3}{15}$$

$$= \frac{-2}{10} - \frac{1}{15} + 0 + \frac{2}{15} + \frac{6}{10} + \frac{9}{15}$$

$$= \frac{10}{15} + \frac{4}{10}$$

$$= \frac{20 + 12}{30}$$

$$= \frac{32}{30}$$

$$E(x) = \frac{16}{15}$$

(3) A random variable X has the probability following distribution.

x :	0	1	2	3	4	5	6	7
$F(X=x)$:	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2+k$

i) Find k:

ii) $P(1.5 < x < 4.5 / x > 2)$

iii) smallest value of λ for which $P(X \leq x) > 1/2$.

Soln:

i) $\Sigma P_i = 1$

$$2k + 0 + k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$$

$$9k + 10k^2 = 1$$

$$10k^2 + 9k - 1 = 0$$

-10

$+10 \quad -1$

9

$$k = \frac{-10}{10}, \frac{1}{10}$$

$$\left(k + \frac{10}{10}\right)\left(k - \frac{1}{10}\right)$$

$$k = -1, \frac{1}{10}$$

we choose $\boxed{k = \frac{1}{10}}$

x :	0	1	2	3	4	5	6	
$F(X=x)$:	0	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{4}{100}$	$\frac{2}{100}$	$\frac{17}{100}$

ii) $P(1.5 < X < 4.5 / x > 2)$

$1.5 \simeq 2$

$4.5 \simeq 5$

$3, 4, 5 \ldots$

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$$= \frac{P((1.5 < x < 4.5) \cap (x > 2))}{P(x > 2)}$$

$2 < x < 5$

$3, 4, 5$ common

$$= \frac{P(x=3) + P(x=4)}{1 - P(x \leq 2)}$$

$$= \frac{\frac{2}{10} + \frac{3}{10}}{1 - \left(\frac{1}{10} + \frac{2}{10}\right)}$$

$$= \frac{5/10}{1 - \frac{3}{10}}$$

$$= \frac{5}{10} \Big/ \frac{7}{10}$$

$$= \frac{5}{7}$$

iii) $P(x \leq \lambda) > \frac{1}{2}$

$\lambda = 0$

$$P(x=0) = 0 \ngtr \frac{1}{2}$$

$\lambda = 1$

$$P(x=1) = \frac{1}{10} \ngtr \frac{1}{2}$$

$\lambda = 2$

$$P(x=2) = \frac{3}{10} \ngtr \frac{1}{2}$$

$\lambda = 3$

$$P(x=3) = \frac{5}{10} \ngtr \frac{1}{2}$$

$\lambda = 4$

$$P(x \leq 4) = \frac{8}{10} > \frac{1}{2}.$$

$\lambda = 4, 5, 6, 7.$

smallest value $\boxed{\lambda = 4}$

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(4) $2P(X=1) = 3P(X=2) = P(X=3) = 5P(X=4)$ ~~[illegible]~~, find pdf & cdf

soln:

$$2P(X=1) = 3P(X=2) = P(X=3) = 5P(X=4) = k \text{ (say)}$$

$$P(X=1) = \frac{k}{2} \; ; \; P(X=2) = \frac{k}{3} \, , \; P(X=3) = k$$

$$P(X=4) = \frac{k}{5}$$

$X=x$	1	2	3	4
$P(X=x)$	$\frac{k}{2}$	$\frac{k}{3}$	k	$\frac{k}{5}$

$$\Sigma P_i = 1$$

$$\frac{k}{2} + \frac{k}{3} + k + \frac{k}{5} = 1$$

$$\frac{15k + 10k + 30k + 6k}{30} = 1$$

$$\frac{61k}{30} = 1$$

$$\boxed{k = \frac{30}{61}}$$

p.m.f of x:

$X=x$	1	2	3	4
$P(X=x)$	$\frac{15}{61}$	$\frac{10}{61}$	$\frac{30}{61}$	$\frac{6}{61}$

c.d.f of x:

$$F(x) = \begin{cases} 0 & -\infty < x < 1 \\ 15/61 & 1 \le x < 2 \\ 25/61 & 2 \le x < 3 \\ 55/61 & 3 \le x < 4 \\ 1 & 4 \le x < 5 \end{cases}$$

(5) The probability function of an infinite discrete distribution is given by $P(X=j) = \frac{1}{2^j}$ where $j = 1, 2, \ldots, \infty$. Verify that the total probability is 1 and find the mean and variance of the distribution. Find also $P(X$ is even$)$, $P(X \geq 5)$, $P(X/3)$.

Soln:-

To find:

$$\sum p_j = 1$$

$$\sum_{j=1}^{\infty} p_j = \frac{1}{2^1} + \frac{1}{2^2} + \frac{1}{2^3} + \ldots + \infty$$

$$= \frac{1}{2}\left[1 + \left(\frac{1}{2}\right)^1 + \left(\frac{1}{2}\right)^2 + \ldots \infty\right]$$

$$= \frac{1}{2}\left[1 - \frac{1}{2}\right]^{-1}$$

$(1-x)^{-1} = 1 + x + x^2 + \ldots + \infty$

$(1-x)^{-2} = 1 + 2x + 3x^2 + \ldots \infty$

$$= \frac{\frac{1}{2}}{\frac{1}{2}}$$

$$= 1$$

Mean:

$$E(x) = \sum x p_i$$

$$E(x) = 1 \times \frac{1}{2} + 2 \times \frac{1}{2^2} + 3 \times \frac{1}{2^3} + \ldots + \infty$$

$$= \frac{1}{2}\left[1 + 2\left(\frac{1}{2}\right) + 3\left(\frac{1}{2}\right)^2 + \ldots + \infty\right]$$

$$= \frac{1}{2}\left[\left(1 - \frac{1}{2}\right)^{-2}\right]$$

$$= \frac{1}{2}\left(\frac{1}{2}\right)^{-2}$$

$$= \frac{1/2}{1/2^2} = 2$$

Variance:

$$V(x) = E(x^2) - [E(x)]^2$$

$$E(x^2) = E[x(x+1) - x]$$

$$= E[x(x+1)] - E(x)$$

$$= 1 \times 2 \times \frac{1}{2} + 2 \times 3 \times \frac{1}{2^2} + 3 \times 4 \times \frac{1}{2^3} + \dots + \infty - 2$$

$$= \frac{2}{2}\left[1 + 3 \times \left(\frac{1}{2}\right) + 6\left(\frac{1}{2}\right)^2 + \dots + \infty\right] - 2$$

$$= \left[1 - \frac{1}{2}\right]^{-3} - 2 \qquad (1-x)^{-3} = 1 + 3x + 6x^2$$

$$= \left[\frac{1}{2}\right]^{-3} - 2$$

$$= [2]^3 - 2$$

$$= 8 - 2$$

$$E(x^2) = 6$$

$$V(x) = 6 - [2]^2$$

$$= 6 - 4$$

$$V(x) = 2$$

$$P(x \text{ is even}) = \frac{1}{2^2} + \frac{1}{2^4} + \frac{1}{2^6} + \dots \infty$$

$$= \frac{1}{2^2}\left[1 + \left(\frac{1}{2^2}\right)^1 + \left(\frac{1}{2^4}\right) + \dots \infty\right]$$

$$= \frac{1}{2^2}\left[1 + \left(\frac{1}{2^2}\right)^1 + \left(\frac{1}{2^2}\right)^2 + \dots \infty\right]$$

$$= \frac{1}{2^2}\left[\left(1 - \frac{1}{2^2}\right)^{-1}\right]$$

$$= \frac{1}{4} \times \frac{4}{3}$$

$$= \frac{1}{3}$$

$$P(x \geq 5) = 1 - P(x < 5)$$

$$= 1 - \left[\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4}\right]$$

$$= 1 - \left[\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16}\right]$$

$$= 1 - \left[\frac{8+4+2+1}{16}\right]$$

$$= 1 - \left[\frac{15}{16}\right]$$

$$= \frac{16-15}{16}$$

$$P(x \geq 5) = 1/16$$

$$P(x/3)$$

$$= \frac{1}{2^3} + \frac{1}{2^6} + \frac{1}{2^9} + \dots \infty$$

$$= \frac{1}{2^3}\left[1 + \frac{1}{2^2} + \frac{1}{2^3} + \dots\right]$$

$$1 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3$$

$$= \frac{1}{8}\left[\left(1 - \frac{1}{2}\right)^{-1}\right]$$

$$= \frac{1}{8} \times \frac{2}{1}$$

$$= \frac{1}{4}$$

Continuous random variable:

1. If the density function of a continuous random variable x is given by $f(x) = \begin{cases} ax, & 0 \leq x \leq 1 \\ a, & 1 \leq x \leq 2 \\ 3a - ax, & 2 \leq x \leq 3 \\ 0, & \text{otherwise} \end{cases}$

i) find a

ii) find cdf of x

iii) find $P(x > 1.5)$

Soln:-

i.) $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$\int_{0}^{3} f(x)\,dx = 1$

$\int_{0}^{1} ax\,dx + \int_{1}^{2} a\,dx + \int_{2}^{3} (3a - ax)\,dx = 1$

$\left[\frac{ax^2}{2}\right]_0^1 + \left[ax\right]_1^2 + \left[3ax - \frac{ax^2}{2}\right]_2^3 = 1$

$\frac{a}{2} + a + \cancel{9a} - \frac{9a}{2} - 6a + \frac{4a}{2} = 1$

$6a - \frac{8a}{2} = 1$

$2a = 1$

$\boxed{a = 1/2}$

$$f(x) = \begin{cases} x/2, & 0 \le x \le 1 \\ \frac{1}{2}, & 1 \le x \le 2 \\ \frac{1}{2}(3-x), & 2 \le x \le 3 \\ 0, & \text{otherwise} \end{cases}$$

ii.) c.d.f of x

$$F(x) = \begin{cases} 0, & -\infty \le x < 0 \\ x^2/4, & 0 \le x < 1 \\ \frac{x}{2} - 1/4, & 1 \le x < 2 \\ & 2 \le x < 3 \\ & 3 \le x < \infty \end{cases}$$

$(x<1)$

$(x<2)$

$(x<3)$

$(x<\infty)$

$$F(x) = P(X<x) = \int_{-\infty}^{x} f(x)\,dx$$

$0 \le x < 1$

$$F(x) = P(X<x) = \int_0^x f(x)\,dx = \int_0^x \frac{x}{2}\,dx = \left[\frac{x^2}{4}\right]_0^x = \frac{x^2}{4}$$

$1 \le x < 2$

$$F(x) = P(X<x) = \int_0^1 f_1(x)\,dx + \int_1^x f(x)\,dx$$

$$= \int_0^1 \frac{x}{2}\,dx + \int_1^x \frac{1}{2}\,dx$$

$$= \left[\frac{x^2}{2}\right]_0^1 + \left[\frac{x}{2}\right]_1^x$$

$$= \frac{1}{4} + \frac{x}{2} - \frac{1}{2}$$

$$= \frac{x}{2} - \frac{1}{4}$$

$2 \le x < 3$:

$$F(x) = \int_0^1 f(x)\,dx + \int_1^2 f(x)\,dx + \int_2^x f(x)\,dx$$

$$= \int_0^1 \frac{x}{2}\,dx + \int_1^2 \frac{1}{2}\,dx + \int_2^x \frac{1}{2}(3-x)\,dx$$

$$= \left(\frac{x^2}{4}\right)_0^1 + \left(\frac{x}{2}\right)_1^2 + \left[\frac{1}{2}\left[3x - \frac{x^2}{2}\right]\right]_2^x$$

$$= \frac{1}{4} + \frac{1}{2} + \frac{1}{2}\left(3x - \frac{x^2}{2}\right) - \frac{1}{2}\left(6 - 4/2\right)$$

$$= \frac{3}{4} + \frac{1}{2}\left(3x - \frac{x^2}{2}\right) - 2$$

$$= -\frac{5}{4} + \frac{1}{2}\left(3x - \frac{x^2}{2}\right)$$

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(1) A random variable X ... If $P(x) = \begin{cases} xe^{-x^2/2}, & x \geq 0 \\ 0, & x < 0 \end{cases}$

a.) show that $P(x)$ is P.d.f

b.) Find its distribution function $P(x)$.

(2) A continuous Random variable x that can assume any value between $x = 2$ & $x = 5$ has a density function given by $f(x) = k(1+x)$. Find $P(x<4)$

(3) A continuous Random variable x has a pdf $f(x) = kx^2e^{-x}$, $(x>0)$ find k, mean & variance.

(1) soln:-

a.) $\int_{-\infty}^{\infty} P(x)\,dx = 1$

$$= \int_{0}^{\infty} x e^{-x^2/2}\,dx$$

let $t = x^2/2$, $dt = dx$

lim $x=0 \Rightarrow t=0$

$x=\infty \Rightarrow t=\infty$

$$= \int_{0}^{\infty} e^{-t}\,dt$$

$$= \left[\frac{e^{-t}}{-1}\right]_0^{\infty}$$

$$= \left[\frac{0-1}{-1}\right]$$

$\int_{-\infty}^{\infty} P(x)\,dx = 1$

$\therefore$ $P(x)$ is P.d.f

b.) c.d.f

$$F(x) = \begin{cases} 0 & -\infty < x < 0 \\ 1-e^{-x^2/2} & 0 \le x < \infty \end{cases}$$

$$F(x) = P(X<x) = \int_0^x f(x)\,dx$$

$$= \int_0^x x e^{-x^2/2}\,dx$$

$t = \frac{+x^2}{2}$ $\quad$ $x=0 \Rightarrow t=0$

$dt = x\,dx$ $\quad$ $x = x \Rightarrow t = \frac{x^2}{2}$

$$= \int_0^{x^2/2} e^{-t}\,dt$$

$$= \left[\frac{e^{-t}}{-1}\right]_0^{x^2/2}$$

$$= \frac{e^{-x^2/2}}{-1} - \frac{e^0}{-1}$$

$$F(x) = 1 - e^{-x^2/2}$$

(2) soln:-

$f(x) = k(1+x), \quad x=2, \; x=5$

i) $P(X<4)$: (Find)

In integral $X \le 4$ $\int^4$ even though there is less than symbol

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_2^5 k(1+x)\,dx = 1.$$

$$k\left[x + \frac{x^2}{2}\right]_2^5 = 1$$

$$k\left[5 + \frac{25}{2} - 2 - \frac{4}{2}\right] = 1$$

$$\boxed{k = \frac{2}{27}}$$

$$f(x) = \frac{2}{27}(1+x), \quad x=2, x=5$$

$$P(x<4) = \int_2^4 f(x)\,dx$$

$$= \frac{2}{27}\left[x + \frac{x^2}{2}\right]_2^4$$

$$= \frac{2}{27}\left[4 + \frac{16}{2} - 2 - \frac{4}{2}\right]$$

$$= \frac{2}{27}(8)$$

$$P(x<4) = \frac{16}{27}$$

5) soln:

$$f(x) = kx^2 e^{-x}$$

$u = x^2$ $dv = e^{-x}dx$

$u' = 2x$ $v_1 = e^{-x}/-1$

$u'' = 2$ $v_2 = e^{-x}/(-1)^2$

$u''' = 0$ $v_3 = e^{-x}/(-1)^3$

To find k: $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$$\int_0^{\infty} kx^2 e^{-x}\,dx = 1$$

$\int u\,dv = uv - u'v_1 + u''v_2 - u'''v_3$

$$k\left[x^2 e^{-x}/(-1) - 2x\frac{e^{-x}}{(-1)^2} + 2\frac{e^{-x}}{(-1)^3}\right]_0^{\infty} = 1$$

$$k\left[0 - \frac{2}{-1}\right] = 1$$

$$\boxed{k = 1/2}$$

$$f(x) = \frac{1}{2}x^2 e^{-x}, \quad x>0$$

Mean:

$$E(x) = \int x f(x)\, dx$$

$E(x) = \begin{cases} \Sigma x P(x) & \text{discrete} \\ \int x f(x) & \text{Continuous} \end{cases}$

$u = x^3$ $\quad dv = e^{-x}dx$

$u' = 3x^2$ $\quad v_1 = \frac{e^{-x}}{-1}$

$u'' = 6x$ $\quad v_1 = \frac{+e^{-x}}{(-1)^2}$

$u''' = 6$ $\quad v_2 = \frac{e^{-x}}{(-1)^3}$

$v_3 = \frac{e^{-x}}{(-1)^4}$

$$= \int_0^\infty \frac{x^3}{2} e^{-x} dx$$

$$= \frac{1}{2}\left[x^3 \frac{e^{-x}}{-1} - \frac{3x^2 e^{-x}}{(-1)^2} + \frac{6x\, e^{-x}}{(-1)^3} - 6\frac{e^{-x}}{(-1)^4}\right]_0^\infty$$

$uv - u'v_1 + u''v_2 - u'''v_3$

$$= \frac{1}{2}\left[0 - \frac{(-6)}{1}\right]$$

$$E(x) = 3$$

Variance:

$$Var(x) = E(x^2) - (E(x))^2$$

$$E(x^2) = \int x^2 f(x)\, dx$$

$$= \frac{1}{2}\int_0^\infty x^4 e^{-x} dx$$

$$= \frac{1}{2}\left[\frac{x^4 e^{-x}}{(-1)} - \frac{4x^3 e^{-x}}{(-1)^2} + \frac{12x^2 e^{-x}}{(-1)^3} - \frac{24x e^{-x}}{(-1)^4} + \frac{24e^{-x}}{(-1)^5}\right]_0^\infty$$

$$= \frac{1}{2}\left[0 - \frac{24}{(-1)}\right]$$

$$E(x^2) = 12$$

$$Var(x) = 12 - (3)^2$$

$$= 12 - 9$$

$$\boxed{Var(x) = 3}$$

(4) A continuous Random variable has a pdf $f(x) = 3x^2$; $0 \le x \le 1$

Find a and b such that

i) $P(X \le a) = P(X > a)$ and

ii) $P(X > b) = 0.05$

(5) The distribution function of Random variable X is given by $F(x) = 1 - (1+x)e^{-x}$; $x \ge 0$. Find the density function, mean and variance of X.

(6) The cdf of a continuous Random variable X is given by

$$F(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \le x < \frac{1}{2} \\ 1 - \frac{3}{25}(3 - x^2), & \frac{1}{2} \le x < 3 \\ 1, & x \ge 3. \end{cases}$$

Find the pdf of X and evaluate $P(|X| \le 1)$ and $P(\frac{1}{3} \le X < 4)$ using both the pdf and cdf.

(4) Soln:-

$$f(x) = 3x^2 ; \quad 0 \le x \le 1$$

i) $P(X \le a) = P(X > a)$

$$\int_0^a f(x)\,dx = \int_a^1 f(x)\,dx$$

$$\int_0^a 3x^2\,dx = \int_a^1 3x^2\,dx$$

$$\left[\frac{3x^3}{3}\right]_0^a = \left[\frac{3x^3}{3}\right]_a^1$$

$$a^3 = 1 - a^3$$

$$2a^3 = 1$$

$a^3 = 1/2$

$a = (1/2)^{1/3}$

ii) $\int_b^1 f(x)\,dx = 0.05$

$\int_b^1 3x^2\,dx = 0.05$

$\left[\frac{3x^3}{3}\right]_b^1 = 0.05$

$1 - b^3 = 0.05$

$b^3 = 1 - 0.05$

$b^3 = 0.95$

$\boxed{b = (0.95)^{1/3}}$

5) soln:-

$F(x) = 1 - (1+x)e^{-x}, \quad x \geq 0$

$F'(x) = f(x)$

$f(x) = 0 - \left\{(1+x)e^{-x}(-1) + e^{-x}(1)\right\}$

$= -\left\{-e^{-x} - xe^{-x} + e^{-x}\right\}$

p.d.f is $f(x) = xe^{-x}, \quad x \geq 0$

Mean $= E(x) = \int_0^\infty x^2 e^{-x}\,dx$

$= \left[x^2\frac{e^{-x}}{(-1)} - 2x\frac{e^{-x}}{(-1)^2} + \frac{2e^{-x}}{(-1)^3}\right]_0^\infty$

$= 2.$

Variance $(x) = E(x^2) - (E(x))^2$

$E(x^2) = \int_0^\infty x^3 e^{-x}\,dx$

$$= \left[\frac{x^3 e^{-x}}{(-1)} - \frac{3x^2 e^{-x}}{(-1)^2} + \frac{6x e^{-x}}{(-1)^3} - \frac{6e^{-x}}{(-1)^4} \right]_0^\infty$$

$$E(x^2) = 6$$

$$Var(x) = 6 - (2)^2$$

$$= 6 - 4$$

$$= 2.$$

H.W

1. Assume that x is a continuous random variable with the following p.d.f

$$f(x) = \begin{cases} A(2x - x^2), & 0 < x < 2 \\ 0, & \text{otherwise} \end{cases}$$

a) What is the value of A?

b) find $P(x > 1)$.

a.)

$$\int_0^2 f(x)\,dx = 1$$

$$A \int_0^2 (2x - x^2)\,dx = 1$$

$$A \left[\frac{2x^2}{2} - \frac{x^3}{3} \right]_0^2 = 1$$

$$A \left[4 - \frac{8}{3} \right] = 1$$

$$A \left[\frac{4}{3} \right] = 1$$

$$\boxed{A = 3/4}$$

$$f(x) = \begin{cases} \frac{3}{4}(2x - x^2), & 0 < x < 2 \\ 0, & \text{otherwise} \end{cases}$$

b.) $P(x>1) = \int_{1}^{2} f(x)\,dx$

$$= \frac{3}{4}\int_{1}^{2} (2x - x^2)\,dx$$

$$= \frac{3}{4}\left[\frac{2x^2}{2} - \frac{x^3}{3}\right]_1^2$$

$$= \frac{3}{4}\left[4 - \frac{8}{3} - 1 + \frac{1}{3}\right]$$

$$= \frac{3}{4}\left[3 - \frac{7}{3}\right]$$

$$= \frac{3}{4}\left[\frac{2}{3}\right]$$

$$P(x>1) = \frac{1}{2}$$

(2.) The CDF of the random variable x is defined by

$$F(x) = \begin{cases} 0 & , \quad x < 2 \\ A(x-2) & , \quad 2 \le x < 6 \\ 1 & , \quad x \ge 6 \end{cases}$$

a.) what is the value of A?

b.) with the above value of A, what is $P(x>4)$?

c.) With the above value of A, what is $P(3 \le x \le 5)$?

a) To find A:

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_{2}^{6} -2A\,dx = 1$$

$$A[-2x]_2^6 = 1$$

$$A[-12+4] = 1$$

$$A[-8] = 1$$

$$\boxed{A = -1/8}$$

$$\therefore f(x) = \begin{cases} 0, & x < 2 \\ 1/4, & 2 \le x < 6 \\ 0, & x \ge 6 \end{cases}$$

b) $P(x>4)$

$$= \int_4^6 f(x)\,dx = \int_4^6 1/4\,dx$$

$$= 1/4\,[x]_4^6$$

$$= 1/4\,[6-4]$$

$$= 1/4\,(2)$$

$$\boxed{P(x>4) = 1/2}$$

c) $P(3 \le x \le 5) = \int_3^5 f(x)\,dx \Rightarrow \int_3^5 1/4\,dx$

$$= 1/4\,[x]_3^5$$

$$= 1/4\,[5-3] = 1/4\,[2]$$

$$P(3 \le x \le 5) = 1/2$$

3. If $P(x) = \begin{cases} x/15, & x = 1,2,3,4,5 \\ 0, & \text{elsewhere} \end{cases}$ find

i) $P(x = 1 \text{ (or) } 2)$ &

ii) $P(1/2 < x < 5/2 \,/\, x > 1)$

G.T:

$$P(x) = \begin{cases} x/15, & x = 1,2,3,4,5 \\ 0, & \text{elsewhere} \end{cases}$$

Soln:

x	1	2	3	4	5
$P(X=x)$	$1/15$	$2/15$	$3/15$	$4/15$	$5/15$

i) $P(X = 1 \text{ or } 2)$

$$P(X=1) + P(X=2)$$
$$= 1/15 + 2/15 = 3/15$$
$$P(X = 1 \text{ or } 2) = 1/5$$

ii) $P\left[\dfrac{1/2 < x < 5/2}{X > 1}\right]$

Note:

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$$\therefore P\left[\frac{1/2 < x < 5/2}{x > 1}\right] = P\left[\frac{(1/2 < x < 5/2) \cap x > 1}{P(x > 1)}\right]$$

$$= \frac{P[1 < x < 5/2]}{1 - P(X \leq 1)}$$

$$= \frac{P(X=2)}{1-P(X=1} \Rightarrow \cancel{X\, P(X}$$

$$= \frac{2/15}{1-1/15} = \frac{2/15}{\frac{15-1}{15}} = 2/14$$

$$P\left[\frac{1/2 < X < 5/2}{X > 1}\right] = 1/7$$

(4) A discrete Random Variable 'x' has the following Probability distribution.

x	0	1	2	3	4	5	6	7	8
$P(x)$	0	3a	5a	7a	9a	11a	13a	15a	17a

Find the value of a, $P(x>3)$, Variance & distribution func of x.

Soln:

To find a: $\Sigma P_i = 1$

$0 + 3a + 5a + 7a + 9a + 11a + 13a + 15a + 17a = 1$

$81a = 1$

$$\boxed{a = 1/81}$$

x	0	1	2	3	4	5	6	7	8
$P(x)$	$1/81$	$3/81$	$5/81$	$7/81$	$9/81$	$11/81$	$13/81$	$15/81$	$17/81$

$P(X<3)$

$= P(X=0) + P(X=1) + P(X=2)$

$= 1/81 + 3/81 + 5/81 \Rightarrow 9/81$

$$\boxed{P(X<3) = 1/9}$$

Variance:

$$Var(x) = E(x^2) - [E(x)]^2$$

$$E(x) = \Sigma x P(x)$$

$$= 3/81 + 6/81 + 21/81 + 36/81 + 55/81 + 78/81 + 105/81 + \frac{136}{81}$$

$$= \frac{444}{81} = 5.5$$

$$E(x^2) = 3/81 + 20/81 + 63/81 + 144/81 + 275/81 + 468/81 + \frac{735}{81} + \frac{1088}{81}$$

$$= \frac{2796}{81} = 34.518$$

$$Var(x) = \frac{2796}{81} - \left[\frac{444}{81}\right]^2$$

$$= \frac{2796}{81} - 30.25$$

$$= \frac{34575}{860} = 4.2685!$$

Cummulative distribution function:

$$F(x) = \begin{cases} 0, & \infty < x < 0 \\ 1/81, & 0 < x < 1 \\ 4/81, & 1 \le x < 2 \\ 9/81, & 2 \le x < 3 \\ 16/81, & 3 \le x < 4 \\ 25/81, & 4 \le x < 5 \\ 36/81, & 5 \le x < 6 \\ 46/81, & 6 \le x < 7 \\ 64/81, & 7 \le x < 8 \\ 1, & 8 \le x < \infty. \end{cases}$$

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(6) soln:-

$$F(x) = \begin{cases} 0 & , x<0 \\ x^2/3 & , 0 \le x < \frac{1}{2} \\ 1 - \frac{3}{25}(3-x)^2 & , \frac{1}{2} \le x < 3 \\ 1 & , x \ge 3 \end{cases}$$

$$f(x) = F'(x) = \begin{cases} 0 & , x<0 \\ 2x & , 0 \le x < 1/2 \\ \frac{6(3-x)}{25} & , \frac{1}{2} \le x < 3 \\ 0 & , x \ge 3 \end{cases}$$

cdf

$$P(|x| \le 1) = P(-1 \le x \le 1) \text{ (or) } \int_{-1}^{1} f(x)\,dx = \int_0^{1/2} + \int_{1/2}^{1}$$

$$= F(1) - F(-1)$$

$$= 1 - \frac{3}{25}(3-1)^2 - (0)$$

$$= 1 - \frac{12}{25}$$

$$P(|x| \le 1) = \frac{13}{25} = 0.52$$

cdf

$$P\left(\frac{1}{3} \le x < 4\right) = F(4) - F\left(\frac{1}{3}\right)$$

$$= 1 - \left(\frac{1}{3}\right)^2$$

$$= 1 - \frac{1}{9}$$

$$= \frac{8}{9} = 0.888$$

cdf = F(x)

pdf = f(x)

Pdf

$$P(|x|\leq 1) = \int_{-1}^{1} f(x)\,dx$$

$$= \int_{0}^{1/2} f(x)\,dx + \int_{1/2}^{1} f(x)\,dx$$

$$= \int_{0}^{1/2} 2x\,dx + \int_{1/2}^{1} \frac{6}{25}(3-x)\,dx$$

$$= \left[x^2\right]_0^{1/2} + \frac{6}{25}\left[3x - \frac{x^2}{2}\right]_{1/2}^{1}$$

$$= \frac{1}{4} + \frac{6}{25}\left[3 - \frac{1}{2} - \frac{3}{2} + \frac{1}{8}\right]$$

$$= \frac{1}{4} + \frac{6}{25}\left[3 - \frac{4}{2} + \frac{1}{8}\right]$$

$$= \frac{1}{4} + \frac{6}{25}\left[\frac{24-16+1}{8}\right]$$

$$= \frac{1}{4} + \frac{\cancel{6}^{3}}{25}\left[\frac{9}{\cancel{8}_{4}}\right]$$

$$= \frac{1}{4} + \frac{27}{100}$$

$$= \frac{100+27}{400} = \frac{127}{400} = 0.3$$

pdf

$$P\left(\frac{1}{3} \le x < 4\right) = \int_{1/3}^{1/2} 2x\,dx + \int_{1/2}^{3} \frac{6}{25}(3-x)\,dx$$

$$= \left[x^2\right]_{1/3}^{1/2} + \frac{6}{25}\left[3x - \frac{x^2}{2}\right]_{1/2}^{3}$$

$$= \frac{1}{4} - \frac{1}{9} + \frac{6}{25}\left(9 - \frac{9}{2} - \frac{3}{2} + \frac{1}{8}\right)$$

$$= 0.8888$$

Definition of Expectation:

Expectation (x)

The mean | expected value of a discrete continuous random variable is

$$E(x) = \bar{x} = \begin{cases} \sum x P(x) & , x \text{ is discrete} \\ \int_{-\infty}^{\infty} x f(x)\,dx & , x \text{ is continuous} \end{cases}$$

1. Find the expected value of the discrete random variable X with the probability mass function

$$P(X=x) = \begin{cases} 1/3 & ; x=0 \\ 2/3 & , x=2 \end{cases}$$

soln:-

x	0	2
$P(X=x)$	$1/3$	$2/3$

$$E(X) = \sum x p(x)$$

$$= 0 \times 1/3 + 2 \times 2/3$$

$$= 4/3$$

Note:

If X and Y are random variable and a, b are constants, then

i) $E(a) = a$

ii) $E(aX) = aE(X)$

iii) $E(aX+b) = aE(X)+b$

iv) $E(X+Y) = E(X)+E(Y)$ [Additive theorem]

v) $Var[X] = E[X^2] - [E[X]]^2$

vi) $Var[aX+b] = a^2 Var[X]$

Moments:

Definition:

The n^{th} moment about origin of random variable X is defined as the expected value of the n^{th} power of X.

For a discrete random variable X, the n^{th} moment is defined as

$$E[X^n] = \sum_i x_i^n p_i = \mu'_n, \quad n \geq 1$$

For a continuous variable X, the n^{th} moment is defined as

$$E[X^n] = \int_{-\infty}^{\infty} x^n f(x)\, dx = \mu'_n$$

Definition:

The n^{th} central moment of a discrete random variable X is its moment about its mean $\bar{X}$ and is defined as

$$E\left[(x-\bar{x}^n)\right] = \sum_i (x_i - \bar{x})^n p_i = \mu_n$$

the n^{th} central moment of a continuous random variable x,

$$E\left[(x-\bar{x})^n\right] = \int_{-\infty}^{\infty} (x-\bar{x})^n f(x)\, dx = \mu_n$$

Relation between moments about origin and moments about mean:

i) $\mu_1 = 0$

ii) $\mu_2 = \mu_2' - (\mu_1')^2$

iii) $\mu_3 = \mu_3' - 3\mu_2' \mu_1' + 6\mu_2' (\mu_1')^2 - 3(\mu_1')^4$

Definition:

Moment generating function (MGF) of a random variable x about the origin is denoted by

$$M_x(t) = E\left[e^{tx}\right] = \begin{cases} \sum_x e^{tx} p_x \text{; if } x \text{ is discrete random variable} \\ \int_{-\infty}^{\infty} e^{tx} f(x)\,dx \text{; if } x \text{ is continuous random variable} \end{cases}$$

$$M_x(t) = E\left[e^{tx}\right] = E\left[1 + \frac{tx}{1!} + \frac{t^2x^2}{2!} + \dots + \frac{t^n x^n}{n!} + \dots\right]$$

$$= 1 + tE(x) + \frac{t^2}{2!}E(x^2) + \frac{t^3}{3!}E(x^3) + \dots + \frac{t^n}{3!}E(x^n) + \dots]$$

$$M_X(t) = 1 + t\mu_1' + \frac{t^2}{2!}\mu_2' + \frac{t^3}{3!}\mu_3' + \dots + \frac{t^n}{n!}\mu_n' + \dots \}$$

Note:

1) μ_1' - coefficient of t in the expansion of $M_X(t)$

μ_2' - coefficient of $\frac{t^2}{2!}$ in the expansion of $M_X(t)$

μ_3' - coefficient of $\frac{t^3}{3!}$ in the expansion of $M_X(t)$

μ_n' - coefficient of $\frac{t^n}{n!}$ in the expansion of $M_X(t)$

ii) Differentiating (1) with respect to t, n times from $t=0$

$$\mu_n' = \frac{d^n}{dt^n}\left[M_X(t)\right]_{t=0}$$

iii) $\mu_1' \rightarrow$ Mean of X

$\mu_2' \rightarrow$ ~~Variance of~~ $E(X^2)$

$\mu_2 \rightarrow$ variance of X

(1) If a random variable X has the MGF $M_X(t) = \frac{3}{3-t}$. Obtain the standard deviation of X.

Soln:

$$M_X(t) = \frac{3}{3-t}$$

$$= \frac{\cancel{3}}{\cancel{3}\left(1-\frac{t}{3}\right)}$$

$$= \left(1-\frac{t}{3}\right)^{-1}$$

$$M_X(t) = 1 + \frac{t}{3} + \left(\frac{t}{3}\right)^2 + \frac{2}{2}\left(\frac{t}{3}\right)^3 + \dots$$

$\mu_1' =$ coeff of t

$= \frac{1}{3} =$ Mean.

$\mu_2' =$ coff of $\frac{t^2}{2!} = \frac{t^2}{2}$

$\mu_2' = \frac{2}{9}$

$\mu_1' = E(x)$

$\mu_2' = E(x^2)$

$var(x) = E(x^2) - (E(x))^2$

$= \mu_2' - (\mu_1')^2$

$= \frac{2}{9} - \left(\frac{1}{3}\right)^2$

$= \frac{2}{9} - \frac{1}{9}$

$Var\ X = \frac{1}{9}$

(2) If X represents the outcome when a fair die is tossed find MGF(x) and hence find E(x) and var(x).

soln:

$p_i = \frac{1}{6}$; $i = 1, 2, 3, 4, 5, 6$

$X(i)$:	1	2	3	4	5	6
$P(X=x)$: $(X=i)$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$

$$MGF = M_X(t) = E\left[e^{tX}\right]$$

$$= \sum_{i=1}^{6} e^{ti} p_i$$

$$M_X(t) = \frac{1}{6}\left[e^t + e^{2t} + e^{3t} + e^{4t} + e^{5t} + e^{6t}\right]$$

$$\mu_n' = \frac{d^n}{dt^n}\left[M_X(t)\right]$$

$$\mu_1' = \frac{d}{dt}\left[M_X(t)\right]_{t=0}$$

$$= \frac{1}{6}\left[e^t + 2e^{2t} + 3e^{3t} + 4e^{4t} + 5e^{5t} + 6e^{6t}\right]_{t=0}$$

$$= \frac{1}{6}\left[1+2+3+4+5+6\right]$$

$$= \frac{1}{6}\left[21\right]$$

$$\mu_1' = \frac{7}{2}$$

$$\mu_2' = \frac{d^2}{dt^2}\left[M_X(t)\right]$$

$$= \frac{1}{6}\left[e^t + 4e^{2t} + 9e^{3t} + 16e^{4t} + 25e^{5t} + 36e^{6t}\right]_{t=0}$$

$$= \frac{1}{6}\left[1+4+9+16+25+36\right]$$

$$= \frac{91}{6}$$

$$Var(X) = \mu_2' - (\mu_1')^2$$

$$= \frac{91}{6} - \left(\frac{7}{2}\right)^2$$

$$= \frac{91}{6} - \frac{49}{4}$$

$$= \frac{364 - 294}{24}$$

$$= \frac{70}{24}$$

$$Var(x) = \frac{35}{12}$$

9/12/25

1. Two cards are drawn at random with replacement from a box which contains four cards number 2, 2, 3 and 3 let x denote the sum of the numbers. find the probability distribution of x and also find the mean and variance.

soln:-

The box of no ; 2, 2, 3, 3

$x \Rightarrow$ sum of the numbers

(2, 2), (2, 3), (3, 3), (3, 2)

The prob. distri. of n of x:

x :	4	5	6
$P(x=x)$:	$1/4$ (2,2)	$2/4$ (2,3), (3,2)	$1/4$ (3,3)

Mean $E(x) = \sum_{i=1}^{\infty} x_i p_i$

$$= 4 \times \frac{1}{4} + 5 \times \frac{2}{4} + 6 \times \frac{1}{4}$$

$$= \frac{4+10+6}{4}$$

$$= \frac{20}{4}$$

$$E(x) = 5$$

Variance:

$$Var(x) = E(x^2) - [E(x)]^2$$

$$E(x^2) = \Sigma x^2 p(x)$$

$$= 16 \times \frac{1}{4} + 25 \times \frac{2}{4} + 36 \times \frac{1}{4}$$

$$= 13 + \frac{25}{2}$$

$$E(x^2) = \frac{51}{2}$$

$$Var(x) = \frac{51}{2} - (5)^2$$

$$= \frac{51}{2} - 25$$

$$Var(x) = \frac{1}{2}$$

(2) A random variable X has density function given by $f(x) = \begin{cases} 2e^{-2x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$. Find

i) The MGF (moment generating function)

ii) Find first four moments about the origin

iii) Mean

iv) ~~...~~ Variance.

soln:

i) Find M.G.f

$$M_X(t) = E[e^{tx}]$$

$$= \int_0^\infty e^{tx} f(x) dx$$

$$= \int_0^\infty e^{tx} 2e^{-2x} dx$$

$$= 2\int_0^\infty e^{-(2-t)x} dx$$

$$= 2\left[\frac{e^{-(2-t)x}}{-(2-t)}\right]_0^\infty$$

$$= 2\left[0 - \frac{1}{-(2-t)}\right]$$

$$= 2\left[\frac{1}{2-t}\right]$$

$$M_X(t) = \frac{2}{2-t}$$

ii) Momements about origin:

$$E(x^n) = \mu_n'$$

$$\mu_1' = \text{coeff of } \frac{t}{1!}$$

$$\mu_2' = \text{coeff of } \frac{t^2}{2!}$$

$\vdots$

$$M_X(t) = \frac{2}{2-t} = \frac{2}{2(1-t/2)} = (1-t/2)^{-1}$$

$$= 1 + \frac{t}{2} + \left(\frac{t}{2}\right)^2 + \left(\frac{t}{2}\right)^3 + \left(\frac{t}{2}\right)^4 + \ldots$$

$$M_X(t) = 1 + \frac{t}{1!}\left(\frac{1}{2}\right) + \frac{t^2}{2!}\left(\frac{2!}{4}\right) + \frac{t^3}{3!}\left(\frac{3!}{8}\right) + \frac{t^4}{4!}\left(\frac{4!}{16}\right) + \ldots$$

$$\mu_1' = \frac{1}{2} = E(X) = \text{Mean}$$

$$\mu_2' = \frac{2!}{4} = \frac{2}{4} = \frac{1}{2} = E(x^2)$$

$$\mu_3' = \frac{3!}{8} = \frac{6}{8} = 3/4$$

$$\mu_4' = \frac{4!}{16} = \frac{24}{16} = \frac{3}{2}$$

iii) Mean $E(X) = 1/2$

iv) $Var(X) = E(x^2) - (E(x))^2$

$$= \frac{1}{2} - \frac{1}{4}$$

$$Var(x) = \frac{1}{4}$$

(5) If X is a random variable whose density function is given by $f(x) = \begin{cases} Ae^{-x}, & 0 < x < \infty \\ 0, & \text{otherwise} \end{cases}$. Find

i) The value of A
ii) Mean of X
iii) Variance of X
iv) n^{th} moment about origin
v) First four moment about origin
vi) First four central moments

Soln:-

i) A value:

$$\int_{-\infty}^{\infty} f(x)\, dx = 1$$

$$\int_{0}^{\infty} Ae^{-x}\, dx = 1$$

$$A\left[\frac{e^{-x}}{-1}\right]_0^{\infty} = 1$$

$$A\left[0 - \frac{1}{-1}\right] = 1$$

$$\boxed{A = 1}$$

$$f(x) = \begin{cases} e^{-x}, & 0 < x < \infty \\ 0, & \text{otherwise} \end{cases}$$

ii) Mean:

$$E(X) = \int_0^{\infty} x e^{-x}\, dx$$

$$= \left[\frac{xe^{-x}}{-1} - \frac{(1)e^{-x}}{(-1)^2}\right]_0^{\infty}$$

$$= 0 - (-1)$$

$$E(X) = 1$$

iii.)

$$Var(x) = E(x^2) - [E(x)]^2$$

$$E(x^2) = \int_0^\infty x^2 f(x) dx$$

$$= \int_0^\infty x^2 e^{-x} dx$$

$$= \left[\frac{2x e^{-x}}{-1} - \frac{2 e^{-x}}{(-1)^2} \right]_0^\infty$$

$$E(x^2) = 2$$

$$var(x) = 2 - (1)^2$$

$$var(x) = 1$$

iv) nth moment about origin:

$$\mu'_n = E(x^n) = \int_0^\infty x^n e^{-x} dx$$

$$\int_0^\infty x^{n-1} e^{-x} dx = \Gamma n = (n-1)!$$

$$\mu'_n = n!$$

$$\int_0^\infty x^n e^{-x} dx = n!$$

v.) first four moments about origin:

$$\mu'_1 = 1! = 1$$

$$\mu'_2 = 2! = 2$$

$$\mu'_3 = 3! = 6$$

$$\mu'_4 = 4! = 24$$

vi.) first four central moments:

$$\mu_1 = 0$$

$$\mu_2 = \mu'_2 - (\mu'_1)^2 = \text{variance}$$

$$= 2 - (1)^2$$

$$= 2 - 1$$

$$\mu_2 = 1$$

$\mu_3 = \mu_3' - 3\mu_2' \cdot \mu_1' + 2(\mu_1')^3$

$= 6 - 3 \times 2 \times 1 + 2(1)^3$

$= 6 - 6 + 2$

$\mu_3 = 2$

$\mu_4 = \mu_4' - 3\mu_3' \mu_2' \mu_1' + 6\mu_2'(\mu_1')^2 - 3(\mu_1')^4$

$= 24 - 3(6)(2)(1) + 6(2)(1)^2 - 3(1)^4$

$= 24 - 36 + 12 - 3$

$\mu_4 = 9$

(4) The density function of random variable x is denoted by $f(x) = Ax(2-x)$, $0 \leq x \leq 2$. Find

i) The value of A
ii) Mean
iii) Variance
iv) nth moment about origin

Soln:-

i)

$f(x) = Ax(2-x)$, $0 \leq x \leq 2$

$\int_0^2 f(x)dx = 1$

$A \int_0^2 2x - x^2 \, dx = 1$

$A \left[\frac{2x^2}{2} - \frac{x^3}{3} \right]_0^2 = 1$

$A \left[4 - \frac{8}{3} \right] = 1$

$A \left[\frac{4}{3} \right] = 1$

$\boxed{A = 3/4}$

iv.) n^{th} moment about origin:

$$\mu'_n = E(x^n) = \int_0^2 x^n \frac{3}{4}(2x-x^2)\,dx$$

$$= \frac{3}{4}\int_0^2 (2x^{n+1} - x^{n+2})\,dx$$

$$= \frac{3}{4}\left[\frac{2x^{n+2}}{n+2} - \frac{x^{n+3}}{n+3}\right]_0^2$$

$$= \frac{3}{4}\left[\frac{2 \times 2^{n+2}}{n+2} - \frac{2^{n+3}}{n+3}\right]$$

$$= \frac{3}{4} \times 2^{n+3}\left[\frac{1}{n+2} - \frac{1}{n+3}\right]$$

$$= \frac{3}{4} \times 2^{n+3}\left[\frac{n+3-n-2}{(n+2)(n+3)}\right]$$

$$= \frac{3}{\cancel{4}} \times 2^n \times \cancel{8}^{2}\left[\frac{1}{(n+2)(n+3)}\right]$$

$$\mu'_n = 3 \times 2^{n+1}\left[\frac{1}{(n+2)(n+3)}\right]$$

ii.) Mean:

$$E(x) = \frac{3}{4}\int_0^2 2x - x^2\,dx$$

$$= \frac{3}{4}\left[x^2 - \frac{x^3}{3}\right]_0^2$$

$$= \frac{3}{4}\left[4 - \frac{8}{3}\right]$$

$$= \frac{3}{4}\left[\frac{4}{3}\right]$$

$$E(x) = 1$$

iii) Variance:

$$Var(x) = E(x^2) - [E(x)]^2$$

$$E(x^2) = \frac{3}{4}\int_0^2 x^2(2x - x^2)\,dx$$

$$= \frac{3}{4} \int_0^2 2x^3 - x^4 \, dx$$

$$= \frac{3}{4} \left[\frac{2x^4}{4} - \frac{x^5}{5} \right]_0^2$$

$$= \frac{3}{4} \left[\frac{2 \times 16}{4} - \frac{32}{5} \right]$$

$$= \frac{3}{4} \left[8 - \frac{32}{5} \right]$$

$$= \frac{3}{4} \left[\frac{8}{5} \right]$$

$$E(x^2) = \frac{6}{5}$$

$$\text{var}(x) = \frac{6}{5} - (1)^2$$

$$\text{var}(x) = \frac{1}{5}$$

(5) Find the moment generating function for

$$f(x) = \begin{cases} 2/3, & x = 1 \\ 1/3, & x = 2 \\ 0, & \text{otherwise.} \end{cases}$$

. Also Find mean and variance

soln:-

$$\text{MGF} = M_x(t) = E[e^{tx}]$$

$$= \sum_{i=1}^{2} e^{tx_i} f(x_i)$$

$$= e^{t(1)} \left(\frac{2}{3}\right) + e^{t(2)} \left(\frac{1}{3}\right)$$

$$M_x(t) = \frac{2e^t}{3} + \frac{e^{2t}}{3}$$

Mean:

$$E(x) = \mu'_1 = \frac{d}{dt}\left(M_x(t)\right)_{t=0}$$

$$= \frac{1}{3}\left[2e^t + 2e^{2t}\right]$$

$$E(x) = \frac{4}{3}$$

Variance:

$$Var(x) = E(x^2) - (E(x))^2$$

$$\mu'_2 = \frac{d^2}{dt^2}\left[M_x(t)\right]_{t=0}$$

$$= \frac{1}{3}\left[2e^t + 4e^{2t}\right]_{t=0}$$

$$= \frac{1}{3}[2+4]$$

$$\mu'_2 = 2$$

$$var(x) = 2 - \left(\frac{4}{3}\right)^2$$

$$= \frac{18-16}{9}$$

$$var(x) = \frac{2}{9}$$

Binomial distribution:

Some standard distribution:

The discrete distribution:

1. Binomial distribution
2. Poission distribution
3. geometric distribution

The continuous distribution:

1. Uniform distribution
2. Exponential distribution
3. Normal distribution

any ①② m from this 6.

Binomial Distribution:

A random variable x is said to be follow binomial distribution with parameters n & P if it satisfies the probability mass function

$$P(x) = P(X=x) = nC_x \, p^x q^{n-x} ; \quad x = 0,1,2,3,\ldots$$

$X \Rightarrow$ Binomial random variable

$P \Rightarrow$ probability of success in a single trial

$q \Rightarrow 1-p =$ probability of failure in a single trial

n - number of trials

x - number of success in n-trials.

Mean $= np$

variance $= npq$

MGF $= M_x(t) = (q + pe^t)^n$

$S.D = \sqrt{Var(x)}$

① The mean of the binomial distribution is 5 and standard deviation is 2. Determine the distribution.

Soln:-

mean $= 5$

$\boxed{np = 5}$

$S.D = \sqrt{Var(x)} = 2$

$Var(x) = (2^2) = 4$

$\boxed{npq = 4}$

$5q = 4$

$\boxed{q = \frac{4}{5}}$

$p + q = 1$

$p = 1 - \frac{4}{5}$

$\boxed{p = 1/5}$

$np = 5$

$n\left(\frac{1}{5}\right) = 5$

$\boxed{n = 25}$

Binomial Distribution

$$P(x=x) = nC_x \, p^x q^{n-x}$$

$$P(x=x) = 25C_x \left(\frac{1}{5}\right)^x \left(\frac{4}{5}\right)^{25-x}$$

2) The mean and variance of binomial distribution are 8 and 6. Find $P(x \geq 2)$.

soln:-

$np = 8$

$npq = 6$

$8q = 6$

$q = 6/8$

$\boxed{q = 3/4}$

$p = 1 - 3/4$

$\boxed{p = 1/4}$

$n(1/4) = 8$

$\boxed{n = 32}$

$$P(x \geq 2) = 1 - P(x < 2) \quad 0, 1$$

$$= 1 - [P(x=0) + P(x=1)]$$

$$= 1 - \left\{\left[32C_0 \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^{32}\right] + \left[32C_1 \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^{31}\right]\right\}$$

$$= 1 - \left\{\left(\frac{3}{4}\right)^{32} + 32 \times \frac{1}{4} \times \left(\frac{3}{4}\right)^{31}\right\}$$

$$= 1 - \{0.000100452 + 0.0010714\}$$

$$= 1 - \{0.00114144\}$$

1.00452×10^{-4}

$$P(x \geq 2) = 0.9988$$

3) If the change of running a bus service according to schedule is 0.8. Calculate the probability as A day schedule with ten services.

1. exactly one is late.
2. Atleast one is late.
3. Atleast two is late.

Soln:-

$\boxed{p=0.8}$

$\boxed{n=10}$

$p+q=1$

$q=1-0.8$

$\boxed{q=0.2}$

The binomial distribution,

$P(X=x) = nC_x \, p^x q^{n-x}, \quad x=0,1,2,\ldots$

$= 10C_x \, (0.8)^x (0.2)^{(10-x)}$

4.1984×10^{-6}

i.) $P(X=1) = 10C_1 \, (0.8)^1 (0.2)^9$

$= 10 \times 0.8 \times (0.2)^9$

4.096×10^{-6}

$= 0.000004096$

0.4096×10^{-5}

0.04

0.004

0.0004

0.00004

0.000004

ii.) $P(x \geq 1) = 1 - P(x<1)$

$= 1 - P(x=0)$

$= 1 - ((0.2)^{10})$

$P(x \geq 2) = 0.9999$

1.024×10^{-7}

0.0000001024

iii.) $P(x \geq 2) = 1 - P(x<2)$

$= 1 - \{P(x=0) + P(x=1)\}$

$= 1 - \{(0.2)^{10} + (10 \times (0.8) \times (0.2)^9)\}$

$= 1 - \{0.0000001024 + 0.0000004096\}$

$= 1 - 0.00000041924$

$P(x \geq 2) = 0.9999$

④ If 10 percent of the screws produced by an automatic machine are defective. the Probability that of 20 screws selected at random. There are

i) Exactly 2 defective

ii) Atmost 3 defective

iii) Atleast 2 defective

iv) Between 1 and 3 defectives, inclusive also find

v) Mean

vi) Variance.

⑤ For a binomial distribution the mean is 4 and variance is 2. Find probability of getting

i) Atleast 2 success

ii) Atmost 2 success

iii) find $P(5 \leq X \leq 7)$

⑥ A binomial variance X satisfies the relation $9P(x=4) = P(x=2)$ when $n=6$, Find the parameter P of the binomial distribution.

⑦ If on an average 9 shifts out of 10 arrived safely to the port. Then obtain the mean and Standard deviation of the no. of ships returning safely out of 150 ships.

④ soln:-

$\boxed{n=20}$

$q = 10\%$

$\boxed{q = 1/10}$

$p = 1 - 1/10$

$\boxed{p = 9/10}$

i) $P(X=2) = 20C_2 \times \left(\frac{9}{10}\right)^2 \times \left(\frac{1}{10}\right)^{18}$

$= 190 \times 0.81 \times 1 \times 10^{-18}$

$= 1.539 \times 10^{-16}$

$P(X \leq 3)$ find

ii) $P(X<3) = \left[20C_0 \times \left(\frac{9}{10}\right)^0 \times \left(\frac{1}{10}\right)^{20}\right] + \left[20C_1 \times \left(\frac{9}{10}\right)^1 \times \left(\frac{1}{10}\right)^{19}\right] +$

$\left[20C_2 \times \left(\frac{9}{10}\right)^2 \times \left(\frac{1}{10}\right)^{18}\right]$

$+ \left[20C_3 \times \left(\frac{9}{10}\right)^3 \left(\frac{1}{10}\right)^{17}\right]$

$= \left(\frac{1}{10}\right)^{20} + \left[20 \times \frac{9}{10} \times 1 \times 10^{-19}\right] + \left[190 \times 0.81 \times 1 \times 10^{-18}\right]$

$+ 1140 \times 0.729 \times 1 \times 10^{-17}$

$= 1 \times 10^{-20} + 1.8 \times 10^{-18} + 1.539 \times 10^{-16} + 831.6 \times 10^{-17}$

$P(X<3) = 8.\cancel{47}17 \times 10^{-15}$

iii) $P(X \geq 2) = 1 - P(X<2)$

$= 1 - [P(X=0) + P(X=1) + \cancel{P(X=2)}]$

$= 1 - \cancel{1.539 \times 10^{-16}} \left[1 \times 10^{-20} + 1.8 \times 10^{-18}\right]$

$= 1 - 1.81 \times 10^{-18}$

$P(X \geq 2) = 1$

iv.) $P(1 \le x \le 3) = P(x=1) + P(x=2) + P(x=3)$

$= 1.8 \times 10^{-18} + 1.539 \times 10^{-16} + 1140 \times \left(\frac{9}{10}\right)^3 \times \left(\frac{1}{10}\right)^{17}$

$= 1.8 \times 10^{-18} + 1.539 \times 10^{-16} + 8.3106 \times 10^{-15}$

$P(1 \le x \le 3) = 8.4663 \times 10^{-15}$

v.) Mean $= np = 20\left(\frac{9}{10}\right)$

$np = 18$

vi.) Variance $= npq = 18\left(\frac{1}{10}\right)$

$npq = 1.8$

5. soln:-

$np = 4$

$npq = 2$

$4q = 2$

$\boxed{q = \frac{1}{2}}$

$n\left(\frac{1}{2}\right) = 4$

$\boxed{n = 8}$

$p = 1 - 1/2$

$\boxed{p = \frac{1}{2}}$

i.) $P(x \ge 2) = 1 - (P(x < 2)$

$= 1 - \{P(x=0) + P(x=1)\}$

$= 1 - \left\{\left(\frac{1}{2}\right)^8 + 8 \times \frac{1}{2} \times \left(\frac{1}{2}\right)^7\right\}$

$= 1 - \{3.9062 \times 10^{-3} + 0.03125\}$

$= 1 - \{0.0039062 + 0.03125\}$

$= 1 - 0.0351562$

$P(x \ge 2) = 0.9648.$

ii) $P(X<2) = P(X=0) + P(X=1) + P(X=2)$

$8C_2 \times \left(\frac{1}{2}\right)^2 \times \left(\frac{1}{2}\right)^6$

$P(X<2) = 0.0351 + 0.105 = 0.1401$

iii) $P(5 \le X \le 7) = P(X=5) + P(X=6) + P(X=7)$

$= 8C_5 \times \left(\frac{1}{2}\right)^5 \times \left(\frac{1}{2}\right)^3 + 8C_6 \times \left(\frac{1}{2}\right)^6 \times \left(\frac{1}{2}\right)^2 + 8C_7 \times \left(\frac{1}{2}\right)^7 \times \left(\frac{1}{2}\right)^1$

$= 56 \times \left(\frac{1}{2}\right)^5 \times \left(\frac{1}{2}\right)^3 + 28 \times \left(\frac{1}{2}\right)^6 \times \left(\frac{1}{2}\right)^2 + 8 \times \left(\frac{1}{2}\right)^7 \times \left(\frac{1}{2}\right)^1$

$= \left[\frac{1}{2}\right]^8 [56 + 28 + 8]$

$\cancel{= 0.0546 + 0.1093 + 0.0312}$

$= \frac{1}{2^8} [92]$

$= 0.3593$

6. soln:

$9P(X=4) = P(X=2)$

$n = 6$

$p + q = 1$

$q = 1 - p$

$9\left[6C_4 \times (p)^4 \times (1-p)^2\right] = \left[6C_2 (p)^2 (1-p)^4\right]$

$9\left[15 (p)^4 (1-p)^2\right] = \left[15 (p^2) (1-p)^4\right]$

$9p^2 = (1-p)^2$

$9p^2 = 1 - 2p + p^2$

$8p^2 = 1 - 2p$

$8p^2 + 2p - 1 = 0$

$(p + 1/2)(p - 1/4) = 0$

$p = -\frac{1}{2}, \frac{1}{4}$

$$\boxed{p = \frac{1}{4}}$$

7. Soln:-

$n = 150$

$$\boxed{p = \frac{9}{10}}$$

$q = 1 - \frac{9}{10}$

$$\boxed{q = \frac{1}{10}}$$

mean $= np = 150\left(\frac{9}{10}\right)$

$np = 135$

$$S.D = \sqrt{var(x)} = \sqrt{npq}$$

$$= \sqrt{(\overset{15}{\cancel{150}})\left(\frac{9}{\cancel{10}}\right)\left(\frac{1}{10}\right)}$$

$$= \sqrt{13.5}$$

$$S.D = 3.67423$$

1/12/25

1. Out of 800 families with 4 children each. How many families would be expected to have

i) Two boys and two girls

ii) Atleast one boy

iii) Atleast two girls

iv) Children of both gender. Assume equal probabilities for boys and girls.

Soln:-

x = no. of boys (failure)

$p = \frac{1}{2}, \; q = \frac{1}{2}$

$n = 4$

$N = 800$

$$P(X=x) = 4C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{4-x}$$

$$= 4C_x \left(\frac{1}{2}\right)^4$$

$$P(X=x) = \frac{4C_x}{2^4}$$

i) $P(\text{2 boys and 2 girls}) = P(X=2)$

$$= \frac{4C_2}{2^4}$$

$$= \frac{6}{16}$$

$$= \frac{3}{8}$$

800 families $= \frac{3}{8} \times 800 = 300$

ii) $P(X \geq 1) = 1 - P(X<1)$

$$= 1 - P(X=0)$$

$$= 1 - \frac{4C_0}{2^4}$$

$$= 1 - \frac{1}{16}$$

$$= \frac{15}{16}$$

800 families $= \frac{15}{16} \times 800 = 750$

iii) $P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$

G: 0 1 2 3 4

B: 4 3 2 1 1

atmost

$$= \frac{1}{16} + \frac{4}{16} + \frac{6}{16}$$

$$= \frac{11}{16}$$

800 families $= \frac{11}{16} \times 800 = 550$

iv) $P(X=1)+P(X=2)+P(X=3)$

$$= \frac{4}{16} + \frac{6}{16} + \frac{4}{16}$$

$$= \frac{14}{16} = \frac{7}{8}$$

800 families $= \frac{7}{8} \times 800 = 700$

Both gender

0	1	2	3	4
4	3	2	1	0

2) It has been founded 80 percentage of the printers used on home computers operate correctly at the time of installation. A particular dealer sales 10 units during a given month

i) find the probability that atleast 9 printers operate correctly on installation

~~ii) ...~~

ii) What is the probability that atleast 9 units operates correctly in each of 5 months.

soln:-

$$P = \frac{80}{100} = 0.8$$

$$q = \frac{20}{100} = 0.2$$

$n = 10$

$$P(X=x) = 10C_x (p)^x (q)^{10-x}$$

$$= 10C_x (0.8)^x (0.2)^{10-x}$$

i) $P(X \geq 9) = P(X=9) + P(X=10)$

$$= 10C_9 (0.8)^9 (0.2)^1 + 10C_{10} (0.8)^{10} (0.2)^0$$

$= 0.2684 + 0.1073$

$= 0.3757$

ii) $(0.3757)^5 = 0.00748$

⑤ There are rainfall in a certain place of 10 days in every 30 days. Find the probability that

i) There ~~is~~ rainfall on atleast 3 days of a given week.

ii) The first four days of a given week will be wet and the remaining days dry.

<u>soln:-</u>

$p = \frac{10}{30} = \frac{1}{3}$ $p+q=1$

$q = 1 - \frac{1}{3}$

$q = \frac{2}{3}$

$N = 30$

$n = 7$

i) $P(x \geq 3) = 1 - P(x<3)$

$= 1 - \{P(x=0) + P(x=1) + P(x=2)\}$

$= 1 - \{7C_0 \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^7 + 7C_1 \left(\frac{1}{3}\right)^1 \left(\frac{2}{3}\right)^6 + 7C_2 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^5\}$

$= 1 - \{0.0585 + 0.2048 + 0.3072\}$

$= 0.4295$

iii) $= \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^3$

$= (0.0123)(0.2962)$

$= 0.00365$

Poisson Distribution:

Note:

i) n, the number of trials is indefinitely large

(i.e) $n \to \infty$

ii) p, constant probability of success for each trial is indefinitely small.

(i.e) $p \to 0$.

Definition:

A random variable x is said to be follow a Poisson distribution if it assumes only non-negative values and its probability mass function is given by

$$P(x=x) = \frac{e^{-\lambda} \cdot \lambda^{x}}{x!} \quad ; x = 0, 1, 2, 3, \ldots$$

$= 0$ otherwise

$\lambda > 0$ (non-negative)

Mean $= \lambda$

Variance $= \lambda$

MGF $= e^{\lambda(e^t - 1)}$

(1) Let x be a random variable following Poisson Distribution such that $P(X=2) = 9P(X=4) + 90P(X=6)$. Find the mean and standard deviation of X.

Soln:-

$$P(X=x) = \frac{e^{-\lambda}\lambda^{x}}{x!}$$

$$P(X=2) = 9P(X=4) + 90P(X=6)$$

$$\frac{e^{-\lambda}\lambda^{2}}{2!} = 9\frac{e^{-\lambda}\lambda^{4}}{4!} + 90\frac{e^{-\lambda}\lambda^{6}}{6!}$$

$$\frac{1}{2} = \frac{9\lambda^{2}}{24} + \frac{90\lambda^{4}}{720}$$

8 16 80 4 240 80 40 8 4

$$4 = 3\lambda^{2} + \lambda^{4}$$

$$[\lambda^{2}]^{2} + 3\lambda^{2} - 4 = 0$$

-4, -1, 4, 3

$$\lambda^{2} = -4, 1$$

$$\lambda^{2} = 1$$

$$\lambda = 1$$

Mean $\lambda = 1$

Variance $\lambda = 1$

$$S.D = \sqrt{var} = \sqrt{1} = 1$$

(2) The no of monthly breakdowns of a computer is a random variable having a Poisson distribution with mean = 1.8. Find the probability that this computer will function for a month

i) without a breakdown

ii) With only one breakdown

iii) Atleast one breakdown

soln:-

$\lambda = 1.8$

$$P(X=x) = \frac{e^{-1.8}(1.8)^x}{x!}, \quad x = 0, 1, 2, \ldots$$

i) $$P(X=0) = \frac{e^{-1.8}(1.8)^0}{0!} = e^{-1.8} = 0.1652$$

ii) $$P(X=1) = \frac{e^{-1.8}(1.8)^1}{1!} = (e^{-1.8})(1.8) = 0.2975$$

iii) $$P(X \geq 1) = 1 - P(X<1)$$

$$= 1 - P(X=0)$$

$$= 1 - 0.1652$$

$$= 0.834$$

Hw:-

① If a poisson variate x is such that $P(X=1) = 2P(X=2)$ Find $P(X=0)$ and variance of x.

② The average no of traffic accidents on a certain town is 2 per week. Assume that the no. of accidents follows a poisson distribution. Find the probability of

$\lambda = 2 \quad P(X=x) = \frac{e^{-2} 2^x}{x!}, \ x = 0, 1, 2.$

i) No accident in a week

$P(X=0)$

ii) Atmost three accident in a two week period $\lambda = 2+2 = 4$

$P(X \leq 3) \quad \frac{e^{-4} 4^x}{x!}$

(3) In a component manufacturing industry there is a small probability of 1/500 for any component to be defective. The components are supplied in packets of 10 use Poisson distribution to calculate the approximate number of packets containing

$p = 1/500$
$n = 10$
$\lambda = np = \frac{1}{50}$

i) No defective
ii) One defective component
iii) Two defective components of 10000 packets.

Soln:-

$$p = 1/500$$

$$n = 10$$

$$\lambda = np = \frac{1}{50} = 0.02$$

$$P(X = x) = \frac{e^{-0.02} (0.02)^x}{x!}, \quad x = 0, 1, 2 \dots$$

i) $P(X = 0) = e^{-0.02} = 0.9801$

ii) $P(X = 1) = e^{-0.02}(0.02) = 0.0196$

iii) $P(X = 2) = \frac{10000 \times e^{-0.02} (0.02)^2}{2!} = 1.9603$

Geometric Distribution:

Definition:

Let X be the number of trails required for the first success in repeated Bernoulli trails with probability of success p, then $P(X=x) = q^{x-1} p, \ x = 1, 2, 3 \dots$

x is called geometric [Pascal] random variable.

H/W

1) soln:-

$$P(x=1) = 2P(x=2) \qquad P(x=x) = \frac{e^{-\lambda}\lambda^{x}}{x!}$$

$$\frac{e^{-\lambda}\lambda^{1}}{1!} = 2\,\frac{e^{-\lambda}\lambda^{2}}{2!}$$

$$\boxed{\lambda=1}$$

Variance = 1

$$P(x=0) = \frac{e^{-1}(1)^{0}}{0!}$$

$$= \frac{e^{-1}}{1}$$

$$P(x=0) = 0.3678$$

2) soln:-

$\lambda=2$ (2 accident per week)

$$P(x=x) = \frac{e^{-2}(2)^{x}}{x!}, \quad x=0,1,2\ldots$$

i) $P(x=0) = \frac{e^{-2}(1)}{1} = 0.1353$

ii) 2 week $\lambda = 2+2 = 4$

$$P(x=x) = \frac{e^{-4}4^{x}}{x!}$$

$$P(x\leq 3) = P(x=0)+P(x=1)+P(x=2)+P(x=3)$$

$$= e^{-4} + \frac{e^{-4}(4)}{1} + \frac{e^{-4}(16)}{2} + \frac{e^{-4}(64)}{6}$$

$$= 0.0183 + 0.0732 + 0.1464 + 0.1952$$

$$P(x\leq 3) = 0.4331.$$

02/1/20

(1) If the probability that an applicant for the driver license will pass the road that any given trial is 0.8. What is the probability that will be finally pass the test? i) On the fourth trial
ii) It fewer than four trial or more than 4 trial.

Soln:-

$$P = 0.8 \qquad q = 1 - 0.8 = 0.2$$

$$P(x = x) = q^{x-1} p \quad x = 1, 2, \ldots \infty$$

$$P(x = x) = (0.2)^{x-1} (0.8)$$

i) $P(x=4) = (0.2)^3 (0.8) = 6.4 \times 10^{-3}$

ii) $P(x>4) = 1 - P(x \le 4)$

$$= 1 - \{P(x=1) + P(x=2) + P(x=3) + P(x=4)\}$$

$$= 1 - \{(0.2)^0 (0.8) + (0.2)^1 (0.8) + (0.2)^2 (0.8) + 6.4 \times 10^{-3}\}$$

$$= 1 - \{0.8 + 0.16 + 0.032 + 6.4 \times 10^{-3}\}$$

$$= 1 - 0.9984$$

$$P(x>4) = 1.6 \times 10^{-3}$$

Uniform Distribution:

Definition:

A continuous random variable x is said to follow a uniform distribution with parameter a b in an interval (a, b). if its probability density function is given by

$$f(x) = \begin{cases} \frac{1}{b-a} & , a < x < b \\ 0 & , \text{otherwise} \end{cases}$$

It is also called as rectangular distribution.

Mean: $E(x) = \mu_1' = \frac{b+a}{2}$

Variance:

$$\mu_2 = \mu_2' - (\mu_1')^2 = \frac{1}{12}(b-a)^2$$

Moment Generating function:

$$M_x(t) = \frac{1}{t(b-a)}\left[e^{tb} - e^{ta}\right], t \neq 0$$

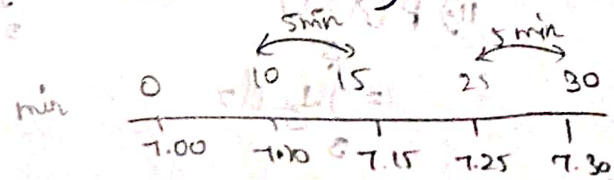
1) Buses arrive at the specify stop at 15 minutes intervals starting at 7 am that is they are at 7, 7:15, 7:30, 7:45 and so on. If a passenger arrives at the stop at a random time that is uniform distributed between 7 and 7.30 am. Find the probability that he waits i) a less than 5 min for a bus.

ii) atleast 12 min for a bus.

soln:-

$$f(x) = \frac{1}{30} \ ; \ 0 < x < 30$$

i) P(less than 5 min)



$$= \int_{10}^{15} f(x)\, dx + \int_{25}^{30} f(x)\, dx$$

$$= \int_{10}^{15} \frac{1}{30}\, dx + \int_{25}^{30} \frac{1}{30}\, dx$$

$$= \frac{1}{30}[x]_{10}^{15} + \frac{1}{30}[x]_{25}^{30}$$

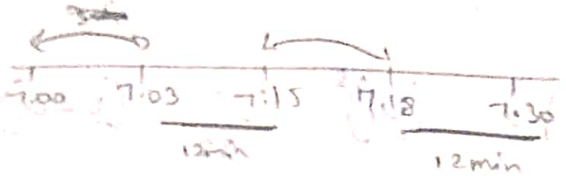
$$= \frac{1}{30}[15-10] + \frac{1}{30}[30-25]$$

$$= \frac{5}{30} + \frac{5}{30}$$

$$= \frac{10}{30}$$

$$= \frac{1}{3}.$$

ii) P (atleast 12 min)



$$= \int_0^3 f(x)\,dx + \int_{15}^{18} f(x)\,dx = \frac{1}{30}[x]_0^3 + \frac{1}{30}[x]_{15}^{18}$$

$$= \frac{(30-0)}{30} + \frac{18-15}{30} = \frac{3+3}{30}$$

$$= \frac{6}{30}$$

$$= \frac{1}{5}$$

P(atleast 12 min) $= \frac{1}{5}$.

(2.) If x is uniformly distributed over 0,5 find

i) $P(x<2)$ ii) $P(x>3)$ iii) $P(2<x<5)$

Soln:- $f(x) = \frac{1}{5} \quad 0<x<5$

i) $P(X<2)$

$$= \int_0^2 \frac{1}{5}\,dx$$

$$= \frac{1}{5}[x]_0^2$$

$$= \frac{1}{5}[2-0]$$

$P(x<2) = \frac{2}{5}$

ii) $P(x>3)$

$$= \int_3^5 \frac{1}{5}\,dx$$

$$= \frac{5-3}{5}$$

$$= \frac{2}{5}$$

iii) $\int_2^5 \frac{1}{5}\,dx$

$$= \frac{5-2}{5}$$

$P(2<x<5) = \frac{3}{5}$

③ Electric trains in a particular route ran every ½ an hour between 12 midnight and 6 am find the probability that a passenger entering the station at any time during this period will have to wait atleast 20 minutes.

Soln:-

$$f(x) = \frac{1}{30}, \quad 0 < x < 30$$

$$30 - 20 = 10$$

$$= \int_0^{10} \frac{1}{30}\,dx$$

$$= \frac{1}{30}(10)$$

$$= \frac{1}{3}$$

④ Let X be random variable with uniformdistribution in the interval $(-a, a)$. Determine a so that i) $P(-1 \le X \le 2) = 0.75$ ii) $P(|X| < 1) = P(|X| > 2)$

Soln:- $(-a, a)$

$$f(x) = \frac{1}{a+a} = \frac{1}{2a}, \quad -a < x < a$$

$$P(-1 \le x \le 2)$$

$$\int_{-1}^{2} \frac{1}{2a}\,dx$$

$$0.75 = \frac{1}{2a}(2+1)$$

$$0.75 = \frac{3}{2a}$$

$$2a = \frac{3}{0.75}$$

$$a = \frac{4}{2}$$

$$\boxed{a = 2}$$

ii) $P(|x|<1) = P(|x|>2)$

$$P(-1<x<1) = P(-2<x<2)$$

$$\int_{-1}^{1} f(x)\,dx = \int_{-2}^{2} f(x)\,dx$$

$$\frac{1}{2a}[x]_{-1}^{1} = \frac{1}{2a}[x]_{-2}^{2}$$

$$\frac{1}{2a}[2] = \frac{1}{2a}[4]$$

$$\frac{1}{a} = \frac{2}{2a}$$

$$2a = a$$

$$2a - a = 0$$

$$\boxed{a=0}$$

(5) Let x be a uniformly distributed random variable (0,1). Determine the moment generating function and hence find the mean & variance.

Soln:

$a=0, b=1$

$$M_x(t) = \frac{1}{t(b-a)}\left[e^{tb} - e^{ta}\right]$$

$$= \frac{1}{t(1-0)}\left[e^{t(1)} - e^{t(0)}\right]$$

$$= \frac{1}{t}\left[e^{t} - 1\right]$$

Mean:

$$E(x) = \mu_1' = \frac{1+0}{2} = \frac{1}{2}$$

Variance:

$\frac{1}{12}(b-a)^2$

$$\mu_2 = \mu_2' - (\mu_1')^2 = \frac{1}{12}(1-0)^2 = \frac{1}{12}$$

Binomial:

(1) A machine manufacturing screws is known to probability 5% defective in a random sample of 15 screws, what is probability that there are

i) exactly 3 defective

ii) not more than 3 defective.

(2) Five fair coins are flipped. if the outcomes are assumed independence find the probability mass function of the number of heads obtained.

(1) soln:-

Probability of five defective screws $p = 0.05$

non-defecting $q = 1-p$
$= 0.95$

$n = 15$

$$P(X=x) = 15C_x (0.05)^x (0.95)^{15-x}$$

i) Exactly 3 defective

$$P(X=3) = 15C_3 (0.05)^3 (0.95)^{12}$$

$$= 455 \times 1.25 \times 10^{-4} \times 0.5403$$

$$= 0.03073.$$

ii) $P(X \leq 3) = P(X=0) + P(X=1) + P(X=2) + P(X=3)$

$$= 15C_0 (0.05)^0 (0.95)^{15} + 15C_1 (0.05)^1 (0.95)^{14} + 15C_2 (0.05)^2 (0.95)^{13} + 15C_3 (0.05)^3 (0.95)^{12}$$

$$= 0.4632 + 0.3657 + 0.1347 + 0.0307$$

$$P(X \leq 3) = 0.9943$$

(2) Soln:-

$n=5$

Each coin two outcomes

Head $= \frac{1}{2} = p$

$q = 1-\frac{1}{2}$

$q=\frac{1}{2}$

$$P(x=x) = 5C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{5-x}$$

Number of heads $x = 0, 1, 2, 3, 4, 5$

$$P(x=x) = P(x=0) = 5C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^5 = \frac{1}{32}$$

$$P(x=1) = 5C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^4 = \frac{5}{32}$$

$$P(x=2) = 5C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^3 = \frac{10}{32}$$

$$P(x=3) = 5C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 = \frac{10}{32}$$

$$P(x=4) = 5C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^1 = \frac{5}{32}$$

$$P(x=5) = 5C_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^0 = \frac{1}{32}$$

6/1/26 **Exponential Distribution:**

A continuous random variable x defined in the interval $(0,\infty)$ is an exponential distribution with parameters $\lambda > 0$ if the probability ~~distribution~~ density function given by

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & , x \geq 0 \\ 0 & , \text{otherwise} \end{cases}$$

Note:

Mean = $1/\lambda$

Variance = $1/\lambda^2$

moment generating function (MGF) = $\frac{\lambda}{\lambda - t}$

Memoryless Property of exponential distribution:

If x is exponentially distributed with parameters λ then for any two positive integer 's' & 't'.

$$P(x > s + t / x > s] = P[x > t)$$

1. The mileage which car owners get with a certain kind of radial is a random variable having an exponential distribution with mean 40,000 km. Find the probability that one of these types

i) atleast 20,000 KM

ii) atmost 30,000 KM

Soln:-

$$\text{Mean} = 40,000 = \frac{1}{\lambda}$$

$$\lambda = \frac{1}{40,000}$$

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$f(x) = \frac{1}{40000} e^{-x/40000}$$

i) $$P(x \geq 20000) = \int_{20000}^{\infty} f(x)\,dx$$

$$= \int_{20000}^{\infty} \frac{e^{-x/40000}}{40000}\,dx$$

$$= \frac{1}{40000}\left[\frac{e^{-\frac{x}{40000}}}{-\frac{1}{40000}}\right]_{20000}^{\infty}$$

$$= \left[\frac{0}{-1} - \frac{e^{-1/2}}{-1}\right]$$

$$P(X \geq 20000) = e^{-1/2}$$

Here x starts from $-\infty$ but we have $x \geq 0$ we put 0

ii) $$P(X \leq 30000) = \int_0^{30000} f(x)\,dx$$

$$= \int_0^{30000} \frac{e^{-\frac{x}{40000}}}{40000}\,dx$$

$$= \frac{1}{\cancel{40000}}\left[\frac{e^{-\frac{x}{40000}}}{\frac{-1}{\cancel{40000}}}\right]_0^{30000}$$

$$= \left[\frac{e^{-3/4}}{-1} - \frac{e^{1}}{-1}\right]$$

$$P(X \leq 30000) = 1 - e^{-3/4}$$

7/1/16

(2) The time required to repair a machine is exponentially distributed with parameter $\lambda = 1/2$.

i) What is probability that repair time exceeds two hours.

ii) What is the conditional probability that a repair takes atleast 10 hour given that its duration exceeds 9 hours.

soln:-

i) $\lambda = 1/2$

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$f(x) = \frac{1}{2} e^{-\frac{1}{2}x}, \quad x \geq 0$$

i) $P(X > 2) = \int_2^\infty f(x)\,dx = \int_2^\infty \frac{1}{2} e^{-\frac{x}{2}}\,dx$

$$= \frac{1}{2}\left[\frac{e^{-x/2}}{-1/2}\right]_2^\infty$$

$$= \frac{1}{2}\left[0 - \frac{e^{-1}}{-1/2}\right]$$

$$= e^{-1}$$

ii) $P(X \geq 10 / X \geq 9) = P(X \geq 9+1 / X > 9)$

$$= P(X > 1)$$

$$= \int_1^\infty \frac{1}{2} e^{-x/2}\,dx$$

$$= \frac{1}{2}\left[\frac{e^{-x/2}}{-1/2}\right]_1^\infty$$

$$= \left[0 - \frac{e^{-1/2}}{-1}\right]$$

$$= e^{-1/2}$$

3) Suppose that during a rainy season in a trophical island. The length of the shower has an exponential distribution with an average 2 minutes. Find the Probability that the shower will be there for more than three minutes, if the shower has already lost for 2 minutes what is the probability that it will lost for atleast 1 more minute.

soln:-

Mean = 2

$\lambda = \frac{1}{2}$

$f(x) = \frac{1}{2} e^{-\frac{1}{2}x}$

i) $P(X > 3) = \int_3^{\infty} \frac{1}{2} e^{-\frac{1}{2}x}$

$= \frac{1}{2} \left[\frac{e^{-\frac{1}{2}x}}{-\frac{1}{2}} \right]_3^{\infty}$

$= \frac{1}{2} \left[0 - \frac{e^{-\frac{3}{2}}}{-\frac{1}{2}} \right]$

$= e^{-3/2}$

ii) $P(X \geq 3 / X > 2) = P(X \geq 2+1 / X > 2)$

$= P(X > 1)$

$= \int_1^{\infty} \frac{1}{2} e^{-x/2} dx$

$= \frac{1}{2} \left[\frac{e^{-x/2}}{-\frac{1}{2}} \right]_1^{\infty}$

$= \frac{1}{2} \left[0 - \frac{e^{-\frac{1}{2}}}{-\frac{1}{2}} \right]$

$= e^{-\frac{1}{2}}$

double check all answers

8M Normal distribution:/z distribution

Def: A random variable x is said to have a normal distribution with parameter μ [called mean] and σ^2 [called variance] if its density function is given by probability law,

$$f(x;\sigma,\mu) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \text{ where } -\infty<x<\infty,$$
$$-\infty<\mu<\infty,$$
$$\sigma>0.$$

Note:

1) A random variable x with mean μ and variance σ^2 and following the normal law is expressed by

$X \sim N(\mu, \sigma^2)$.

2) If $X \sim N(\mu,\sigma^2)$, then $z = \frac{X-\mu}{\sigma}$, is the standard normal variate.

1. If the actual amount of instant coffee which a filling machine puts into six ounce jars is a random variable having a normal distribution with standard deviation 0.05 ounce. and if only 3% of Jars are to contain less than 6 ounce of coffee. what must be the mean fill of the jars.

mean $= \mu$
Variance $= \sigma^2$
$S.D = \sigma$

Soln: Given.

$S.D = \sigma = 0.05$

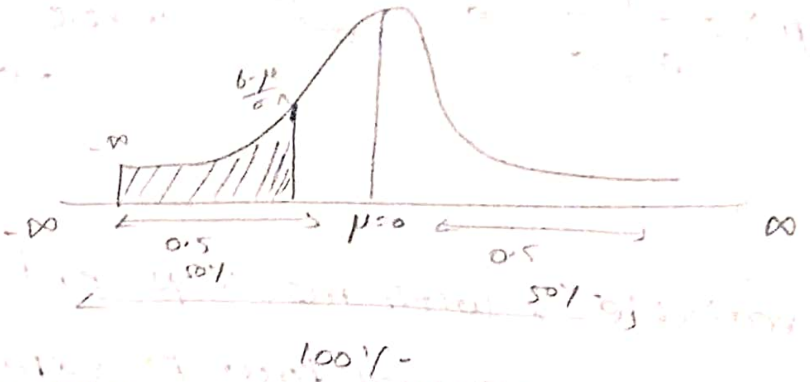
$P(X<6) = 3\%$.

$P(-\infty < X < 6) = 0.03$

$P\left(-\infty < \frac{X-\mu}{\sigma} < \frac{6-\mu}{\sigma}\right) = 0.03$

$P\left(-\infty < Z < \frac{6-\mu}{\sigma}\right) = 0.03$.

$P\left(-\infty < Z < \frac{6-\mu}{0.05}\right) = 0.03$



from the graph

$$0.5 - P\left(0 < z < \frac{6-\mu}{0.05}\right) = 0.03$$

$$0.5 - 0.03 = P\left(0 < z < \frac{6-\mu}{0.05}\right)$$

$$0.47 = P\left(0 < z < \frac{6-\mu}{0.05}\right)$$

from the table

$$0.47 \simeq 0.4699 \qquad (1.8 + 0.08) \to \text{from table}$$

$$0.4699 = P(0 < z < +1.88)$$

$$\frac{6-\mu}{0.05} = -1.88$$

$$6-\mu = -1.88 \times 0.05$$

$$6-\mu = -0.094$$

$$6+0.094 = \mu$$

$$\boxed{\mu = 6.094}$$

2. The marks obtained by a no. of students in a certain subject or approximately normally distributed with mean 65 and S.D 5. if three 3 students are selected at random from this group. what is the probability that atleast

one of them would have score above 75.

Given:

mean = $\mu = 65$

S.D = $\sigma = 5$, $n = 3$

probability of score above 75

$$p(x>75) = p(75<x)$$

$$= p(75<x<\infty)$$

$$= p\left(\frac{75-\mu}{\sigma} < \frac{x-\mu}{\sigma} < \infty\right)$$

$$= p\left(\frac{75-\mu}{\sigma} < z < \infty\right)$$

$$= p\left(\frac{75-65}{5} < z < \infty\right)$$

$$p(x>75) = p(2<z<\infty)$$

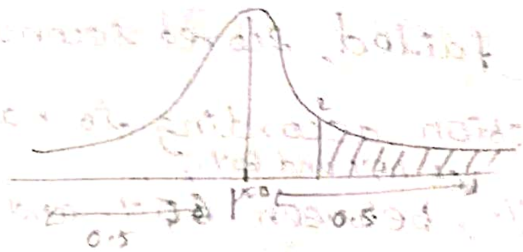


table 0.00

$$= 0.5 - p(0<z<2) \Rightarrow$$

2.00 ⟶ 0.4772

2.00 + 0.00

$$= 0.5 - 0.4772$$

$$P(x>75) = 0.0228$$

Therefore: $p = 0.0228$

$q = 1 - 0.0228 = 0.9772$

given: $n = 3$

P(atleast 1 of them score above 75)

using Binomial distribution.

$$P(x \geq 1) = 1 - P(x < 1)$$

$$1 - P(x = 0)$$

$$= 1 - \left[3c_x (p)^a (q)^{n-x} \right]$$

$$= 1 - \left[3c_0 (p)^a (q)^{3-0} \right]$$

$$= 1 - \left[1 \cdot (0.0228)^0 (0.9772)^3 \right]$$

$$= 1 - 0.9331$$

$$P(x \geq 1) = 0.0668$$

3. In an engineering examination a student considered to have failed, scored second class, first class and distinction according to has he scores ~~less than~~ 45%, bet 45% and 60%, between 60% and 75% and above 75% respectively. In a particular year 10% of the students failed in the examination and 5% of the students got distinction. find the percentages of the students who have got first class and second class [assume normal distribution of marks].

soln

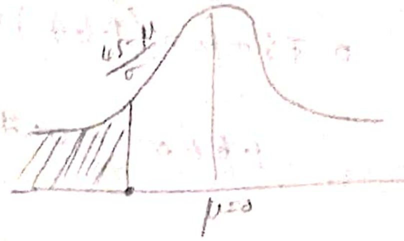
Failure	S.C	F.C	Distinction
<45	$45 < x < 60$	$60 < x < 75$	$x > 75$.
10%			5%.

$p(X < 45\%) = 10\%.$

$p(X < 45\%) = 0.10$

$p(-\infty < X < 45\%) = 0.10$

$p\left(-\infty < Z < \frac{45\% - \mu}{\sigma}\right) = 0.10$



$p(0.5 - P\left(0 < Z < \frac{45-\mu}{\sigma}\right) = 0.10$

$0.5 - P\left(0 < Z < \frac{0.45 - \mu}{\sigma}\right) = 0.10$

$0.40 = P\left(0 < Z < \frac{0.45 - \mu}{\sigma}\right)$

$0.4 \approx 0.3997$ $\qquad 1.2 \longrightarrow 0.3997$ (0.08)

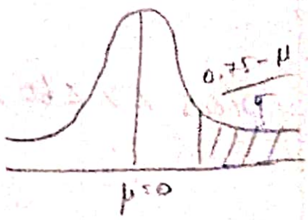
$0.3997 = P\left(0 < Z < \frac{0.45 - \mu}{\sigma}\right)$

$\frac{0.45 - \mu}{\sigma} = -1.28$ because it less than 0

$0.45 - \mu = \sigma(1.28)$

$0.45 = -\sigma(1.28) + \mu \longrightarrow (1)$

$P(X > 75\%) = 5\%.$



$P(0.75 < X < \infty) = 0.05$

$P\left(\frac{0.75 - \mu}{\sigma} < Z < \infty\right) = 0.05$

$0.5 - P\left(\frac{0.75 - \mu}{\sigma} < Z < 0\right) = 0.05$

$0.45 - P\left(0 < Z < \frac{0.75 - \mu}{\sigma}\right).$ $\qquad 0.45$

$0.45 \approx 0.4495$ $\qquad 1.6 \longrightarrow 0.4495$ (0.04)

$$\frac{0.75-\mu}{\sigma} = 1.64$$

$$0.75-\mu = (1.64)\sigma$$

$$1.64\sigma - 0.75 + \mu = 0. \quad \text{—(2)}$$

$$\sigma = 10.27$$

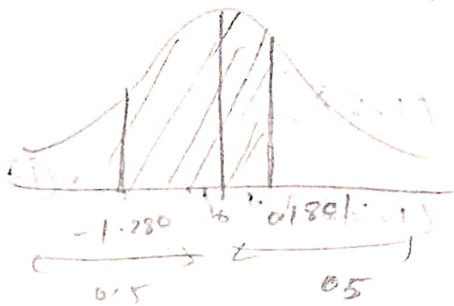
$$\mu = 58.15$$

Second class

$$P(45 < x < 60) = P\left(\frac{45-\mu}{\sigma} < x < \frac{60-\mu}{\sigma}\right)$$

$$= P\left(\frac{45-58.15}{10.27} < x < \frac{60-58.15}{10.27}\right)$$

$$= P(-1.280 < x < 0.1801)$$



$$P(45 < x < 60) = P(-1.280 < z < 0) + P(0 < z < 0.1801)$$

$$= P(0 < z < 1.280) + P(0 < z < 0.1801)$$

0.08
1.2 →
0.08
0.1 →

$$= 0.3997 + 0.0714$$

$$= 0.4711$$

$$= 47.11\% \simeq 47\%$$

First class: $= 100 - (10 + 5 + 47)$

$= 38\%$.

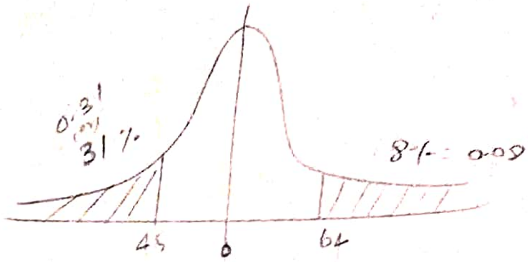
11/16

4. In a normal distribution, 31% of items are under 45 and 8% are over 64. Find the mean and S.D of Distribution.

Soln

(i) $P(x < 45) = 31\%$.

(ii) $P(64 < x < \infty)\ 8\%$.



$P(-\infty < x < 45) = 31\%$. $P(64 < x < \infty) = 8\%$.

$\cancel{0.5} - P$

$$P\left(-\infty < z < \frac{45-\mu}{\sigma}\right) = 0.31$$

$$0.5 - P\left(0 < z < \frac{45-\mu}{\sigma}\right) = 0.31$$

$$0.19 = P\left(0 < z < \frac{45-\mu}{\sigma}\right)$$

0.09
0.4 — 0.1879

$$\frac{45-\mu}{\sigma} = -0.49$$

$$45 - \mu = -0.49\sigma$$

$$-0.49\sigma + \mu = 45 \rightarrow (1)$$

$P(64 < x < \infty) = 8\%$.

$$0.5 - P\left(\frac{64-\mu}{\sigma} < z < \infty\right) = 0.08$$

0.00
1.4 → 0.4192

$$0.42 = P\left(\frac{64-\mu}{\sigma} < z < \infty\right)$$

$$\frac{64-\mu}{\sigma} = 1.4$$

$64 - \mu = 1.4\sigma$

$1.4\sigma + \mu = 64 \longrightarrow (2)$

$\mu = 49.9$

$\mu \simeq 50$

$\sigma = 10$